Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

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Experiment No: 02

Aim: To understand Version Control System / Source Code Management, install git and create a GitHub account

Theory:

- Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. Git is easy to learn and has a tiny footprint with lightning fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.

- Some of the basic operations in Git are:

    1. Initialize
    2. Add
    3. Commit
    4. Pull
    5. Push

- Some advanced Git operations are:
    1. Branching
    2. Merging
    3. Rebasing

The following diagram depict the all supported operations in GIT

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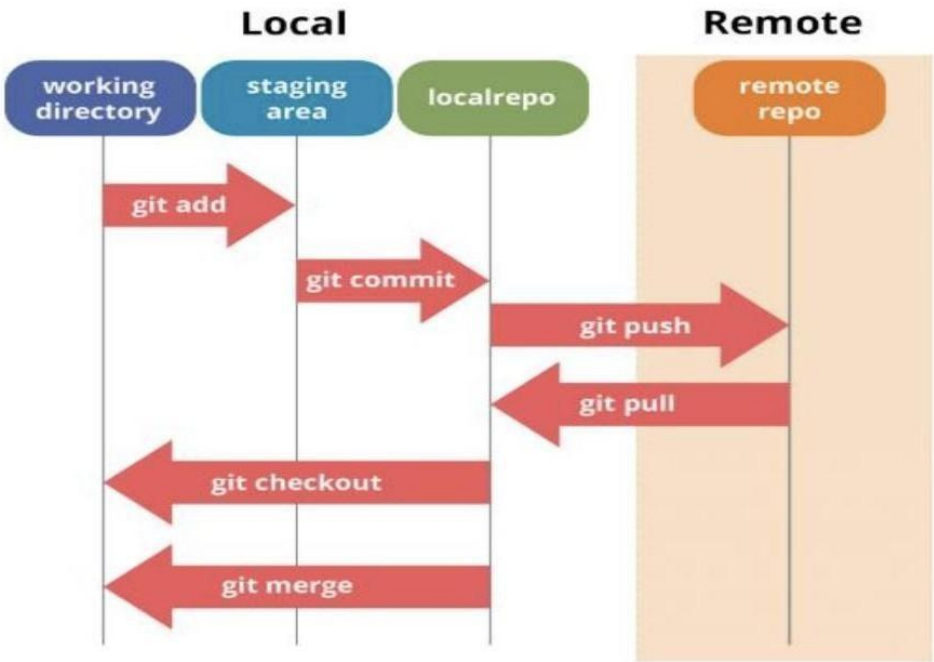
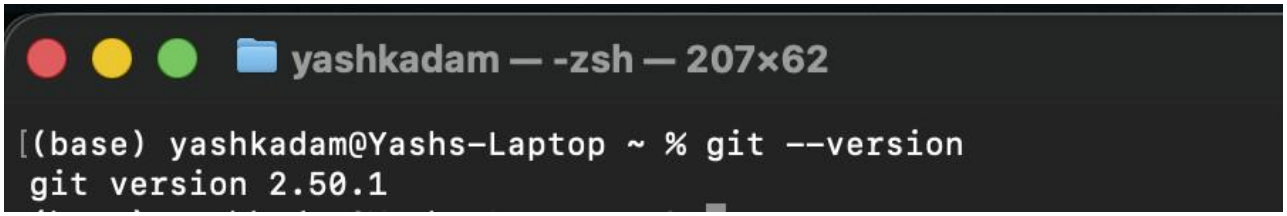


- **Installation of GIT**

1) In windows, download GIT from https://git-scm.com/ and perform the straightforward installation.

2) In Ubuntu, install GIT using $sudo apt install git, Confirm the version after installation using command $git –version



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Output:

```
SEPM_52 — -zsh — 207×62

(base) yashkadam@Yashs-Laptop SEPM_52 % ls
Experiment_02   SEPM_Sem6
(base) yashkadam@Yashs-Laptop SEPM_52 %
```

```
Experiment_02 — -zsh — 207×62

(base) yashkadam@Yashs-Laptop ~ % git --version
git version 2.50.1
(base) yashkadam@Yashs-Laptop ~ % mkdir_52
zsh: command not found: mkdir_52
(base) yashkadam@Yashs-Laptop ~ % mkdir SEPM_52 cd SEPM_52
mkdir: SEPM_52: File exists
(base) yashkadam@Yashs-Laptop ~ % cd SEPM_52
(base) yashkadam@Yashs-Laptop SEPM_52 % ls
(base) yashkadam@Yashs-Laptop SEPM_52 % mkdir Experiment_02
mkdir SEPM_Sem6
(base) yashkadam@Yashs-Laptop SEPM_52 % ls
Experiment_02   SEPM_Sem6
(base) yashkadam@Yashs-Laptop SEPM_52 % cd Experiment_02
git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/yashkadam/SEPM_52/Experiment_02/.git/
(base) yashkadam@Yashs-Laptop Experiment_02 % touch demo.txt
(base) yashkadam@Yashs-Laptop Experiment_02 % git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        demo.txt

nothing added to commit but untracked files present (use "git add" to track)
(base) yashkadam@Yashs-Laptop Experiment_02 % git add demo.txt
(base) yashkadam@Yashs-Laptop Experiment_02 % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   demo.txt

(base) yashkadam@Yashs-Laptop Experiment_02 % git config --global user.name "Yash"
(base) yashkadam@Yashs-Laptop Experiment_02 % git config --global user.email "yash.kadam3605@gmail.com"
(base) yashkadam@Yashs-Laptop Experiment_02 % git commit -m "First commit"
[master (root-commit) 2f1d62d] First commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 demo.txt
(base) yashkadam@Yashs-Laptop Experiment_02 %
```

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Conclusion: Successfully performed Version Control System / Source Code Management, install git and create a GitHub account